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SERVAY ON 'MMWAVE RADAR'

WEEKLY SEMINAR – MIN HO LEE

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BACKGROUND

- Traditionally, robotics systems have relied on light-based sensors, such as cameras and lidar, for building a representation of their environments.
- Unfortunately, cameras and lidars are not functional when encountering illuminationally or structurally degenerate cases.
  - Despite successive efforts on calibration, they suffer degradation in a harsh environment.
- Since at least 2005, mmWave radar has been deployed in vehicles for assisted cruise control and automatic braking systems. Also robotics researchers began utilizing radar in broader contexts.
  - It makes Increased interest for radar in robotics.
  - Benefits of radar can be alternative and complementary sensor.
- To correctly use radar-based systems and understand the measurements, It is **essential** to **understand principle of radar.**

Sensor	Basic Sensor Information				Data Generated			Operation Capability	
	Data Types	Range	Data Density	Resolution	Range	Angle	Velocity	Darkness	VDE
Cameras	2D Images (RGB/BW)	Medium	Dense	High	Yes*	Yes*	No	Limitted	Limitted
Flash Lidar/Depth Sensors	Depth Maps	Short	Dense	High	Yes	Yes	No	Full	Occluded
Lidar	3D Point Clouds	Long	Sparse	High	Yes	Yes	No	Full	Occluded
2D Scanning Radar	Polar Range Map	Long	Dense	High	Yes	Yes	No	Full	Full
3D SoC Radar	Spectral Heatmap	Medium-Long	Dense	Medium	Yes	Yes	Yes	Full	Full
	3D Point Clouds	Long	Very Sparse	Low	Yes	Yes	Yes	Full	Full

Table 1. Basic sensor information of various exteroceptive sensors commonly used in robotics [1]

[1] Harlow, Kyle, et al. "A new wave in robotics: Survey on recent mmwave radar applications in robotics." *arXiv preprint arXiv:2305.01135* (2023).

BASIC PRINCIPLE

- Radar measurement of range, or distance, is made possible because of the properties of radiated **electromagnetic** (EM) energy.
- Electromagnetic waves
  - Electromagnetic energy travels through air at a **constant speed**, at approximately the speed of light ($c_0 = 3 \cdot 10^8\ m/s$)
  - This constant speed allows the determination of the distance between the reflecting objects and the radar.
  - If reflected waves are received, that means an obstacle is in the propagation direction. Thus the direction (in **azimuth** and **elevation**) of the reflecting objects can be measured.

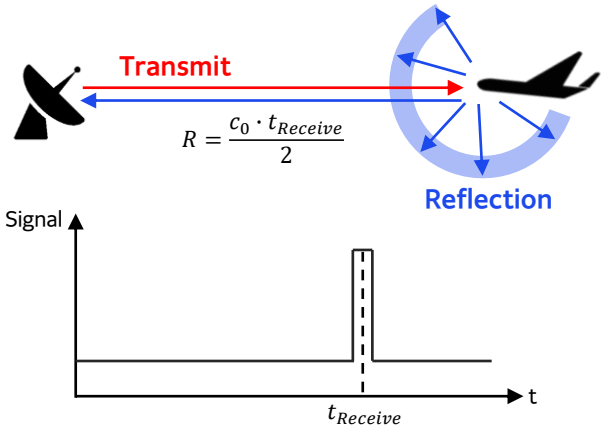


Fig 1. Range determination principle

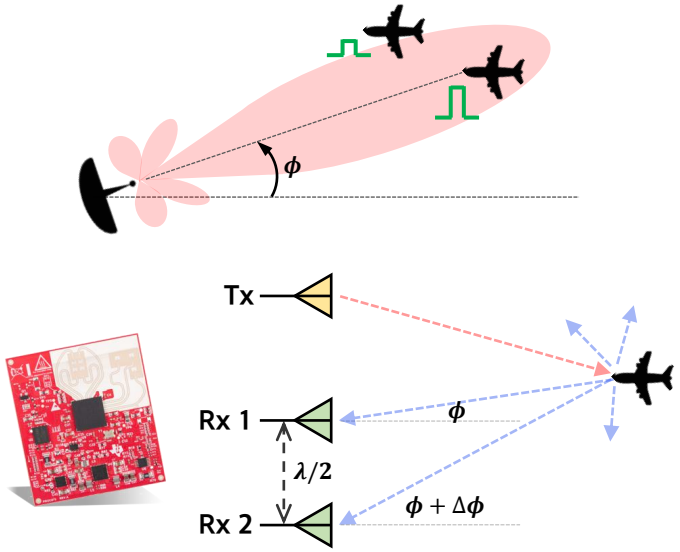
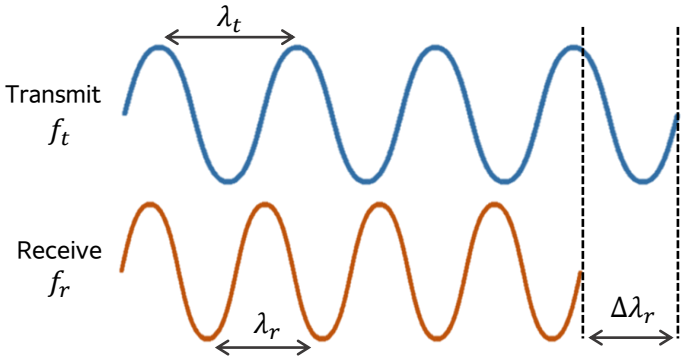
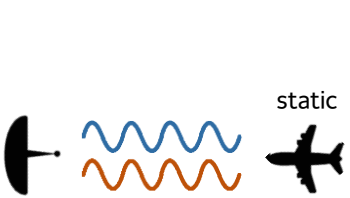


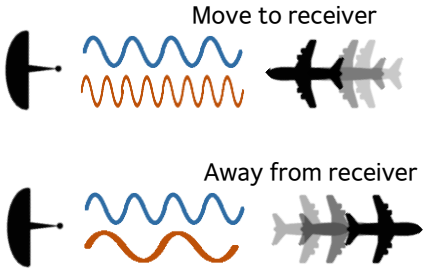
Fig 2. Direction determination principle

BASIC PRINCIPLE

- **Doppler-Effect** is the apparent change in **frequency** when object moves either toward or away from the receiver, or when the receiver moves either toward or away from the object.
  - This principle is used for speed measuring and moving target indication.







$$f_d = f_t - f_r \qquad v_d = c_0 \left(\frac{\lambda_t - \lambda_r}{\lambda_t} \right) \; or \; \frac{c_0 f_d}{2 f_t}$$

- The velocity of the object is calculated using the wavelength of an electromagnetic wave from the radar antenna to the object.
  - If the receiver and object both are moving, the velocity of the target is not only related to target but also receiver. Therefore, the velocity is relative.
  - So, if both direction and velocity of the radar and object are the same, relative velocity is zero.

What is the 'mmWave' Radar?

- 'mmWave' means that electromagnetic **wavelengths** are between 1 and 10mm with frequencies ranging from 30 to 300GHz [1].
  - Applications such as automotive systems using 76-81 GHz and commercial mmWave radar operate in this frequency.
- Short wavelengths make the size of the radar system small and return high-resolution measurements.
- Two types of mmWave radar are commonly used.
  - Radar can generate various types of data format which depends on the antenna types.
  - **SoC** (System On Chip) **Radar**
    - Combine the processing unit into a limited number of chips mounted directly patch antennas, or antenna placed directly into a printed circuit board.
    - Its accuracy and resolutions are dependent on the specific antenna array employed, and the manufacturer-specific processing used to correlate measurements across the antenna.

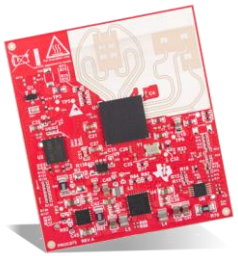


Fig 3. TI SoC Radar

  - **Imaging Radar**
    - It, Sometimes called 'scanning radar', is composed of a physically rotating radar sensor.
    - Imaging radar provides a highly accurate polar image of their surroundings by measuring target ranges around in a series of angles.
    - But, It does not provide 'Doppler Velocity' of the returned targets.



Fig 4. Navtech Scanning Radar

FEATURES OF MMWAVE

- **CW** (Continuous Wave) Radar transmits high-frequency signals continuously.
  - CW radar transmitting an unmodulated power can measure the **speed only** by using the Doppler-Effect.
  - It cannot differentiate between two or more reflecting objects.
- In contrast, **FMCW** (Frequency Modulated Continuous Wave) Radar radiates continuous transmission power with modulated operating frequency.
  - The differences in phase or frequency between the transmitted and the received signal can be calculated.
  - FMCW radar can measure simultaneously target range and its relative velocity.
  - Ability to measure very small ranges to the target and high accuracy of range measurement.

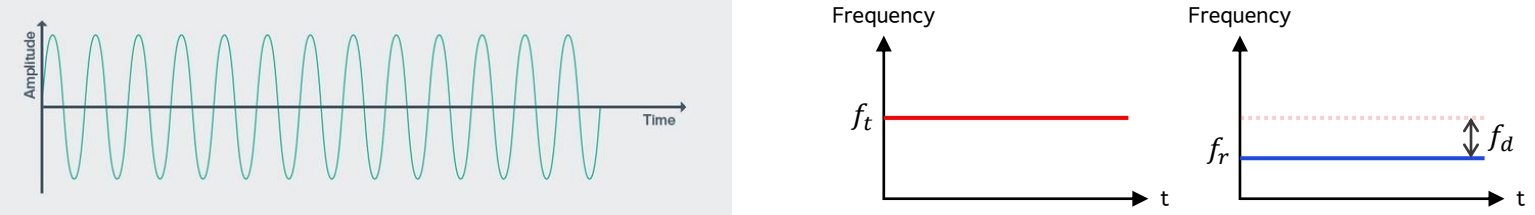


Fig 5. CW shapes with transmit/ received frequency in time domain

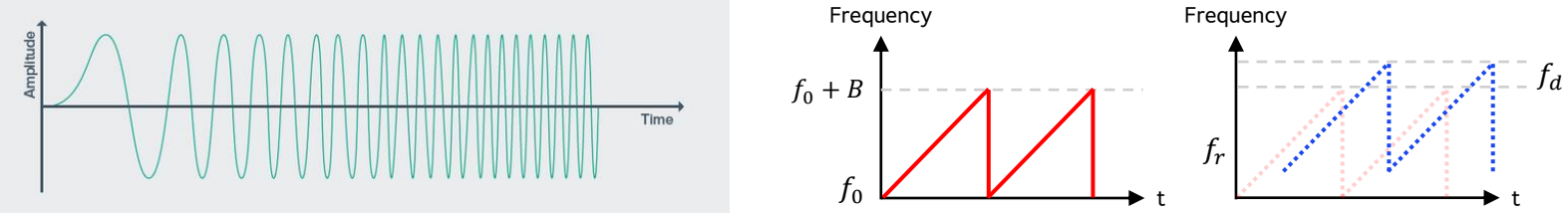


Fig 6. FMCW shapes with transmit/received frequency in time domain

Challenges of Radar system

- Radar returns are produced from nearly all surfaces when illuminated by radar. Therefore, there are unwanted signals returned that involve true objects.
  - These unwanted signals are generally described as **Noise or Clutter**.
  - The definition of clutter depends on the function of the radar.
- Clutter may caused by man-made objects such as buildings and radar countermeasures objects. Other causes include natural objects such as terrain features, sea, birds, turbulence, etc.
- Another spurious return called **Multipath** is derived from detection paths differing for one objects
  - Transmitted rays can reach the receiver directly, However, other rays reach the antenna after being reflected from the ground, after hitting a wall, or others.
- Both clutter and multipath returns degrade the ability of radar

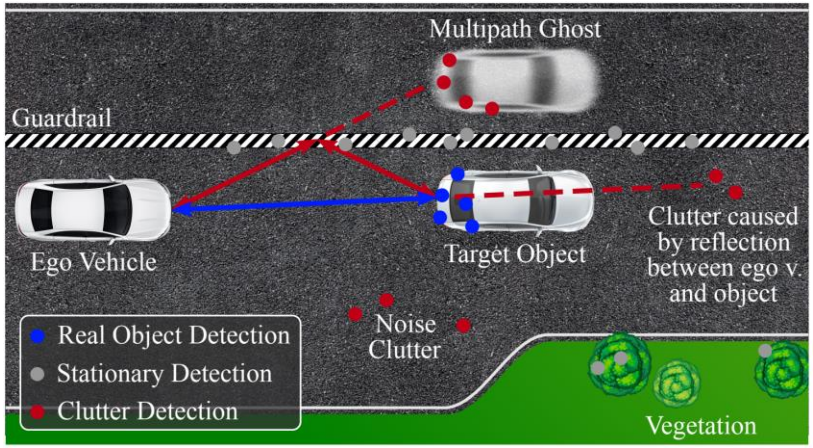


Fig 7. Illustration of multipath caused by variety components

Challenges of Radar system

- **Motion distortion** is an inherent characteristic of scanning sensors including lidar and radar.
- Since the imaging radar generates every single polar image from one rotation, the actual vehicle poses at the first ray and the last ray are different if the vehicle is in motion.
  - The gap between the first and last ray is nonnegligible when the sensor has a low frequency or the vehicle moves at high speed.
- This makes it difficult to accurately detect the position or shape of targets.

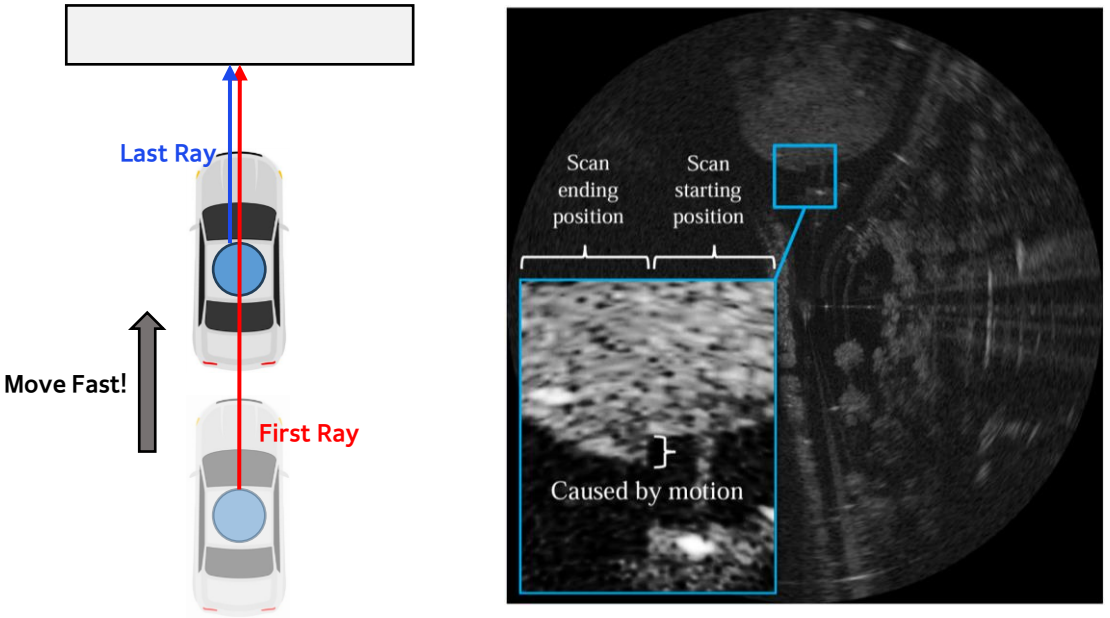


Fig 8. occurred motion distortion in polar image

RADAR FILTERING METHOD - NOISE

- A common method to detect target returns against a background of noise, clutter, and interference is **CFAR** (Constant False Alarm Rate) detector.
- Measurements exceeding the threshold are associated with returns from a target, and measurements below the threshold are associated with noise or other interference.
  - If the threshold is too low, CFAR tends to generate the wrong false alarm.
  - If the threshold is too high, CFAR may fail to detect actual targets.
- Although a constant threshold may work well in noiseless environments, it is difficult to use in noisy environments.
  - A method that is **adaptive** to the environment is required !!

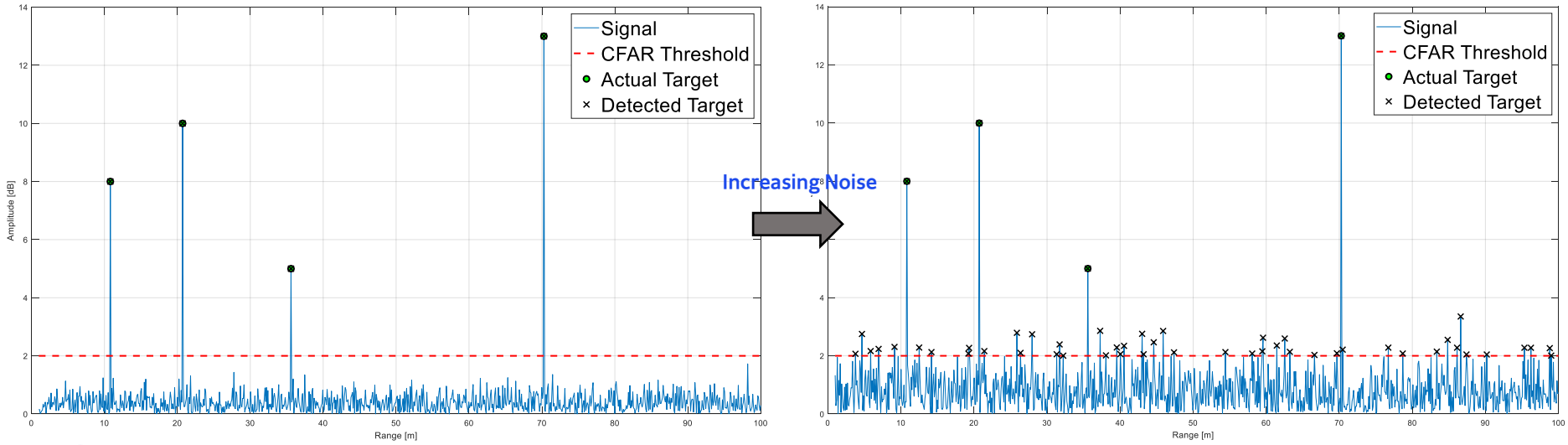
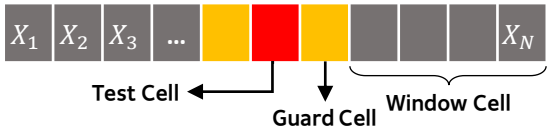


Fig 9. CFAR result using a constant threshold

RADAR FILTERING METHOD - NOISE

- **CA**(Cell-Averaging) CFAR can be used in various environments where the noise level changes.
- The threshold level is calculated by averaging the noise level around the cell window.



$$Threshold = \frac{\alpha}{N}\sum_{i=1}^{N} X_i$$

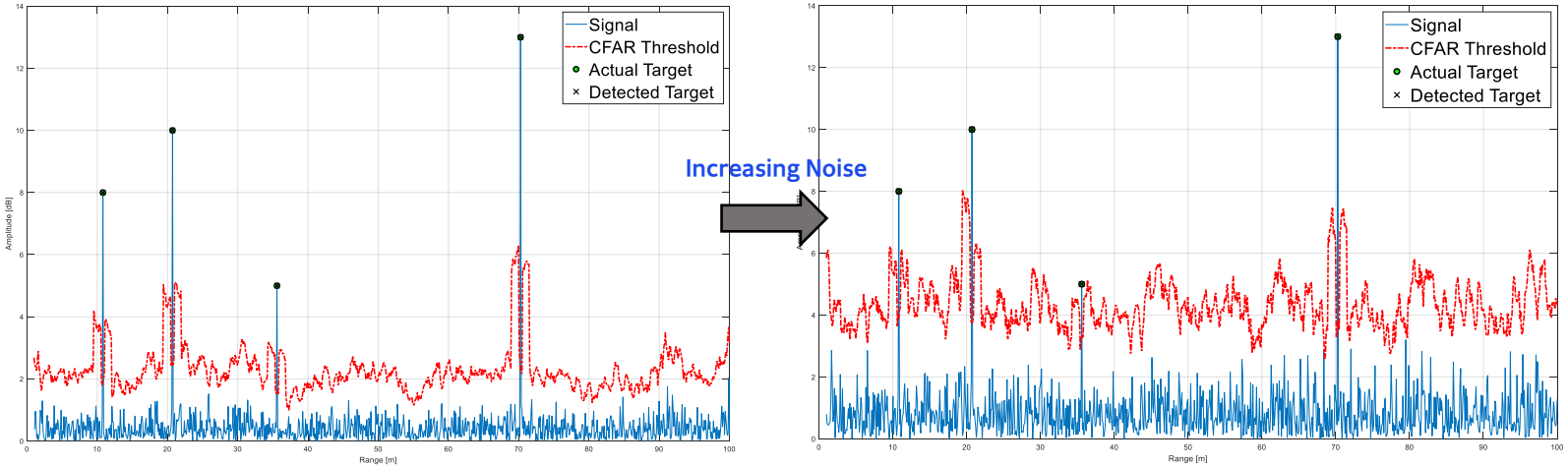
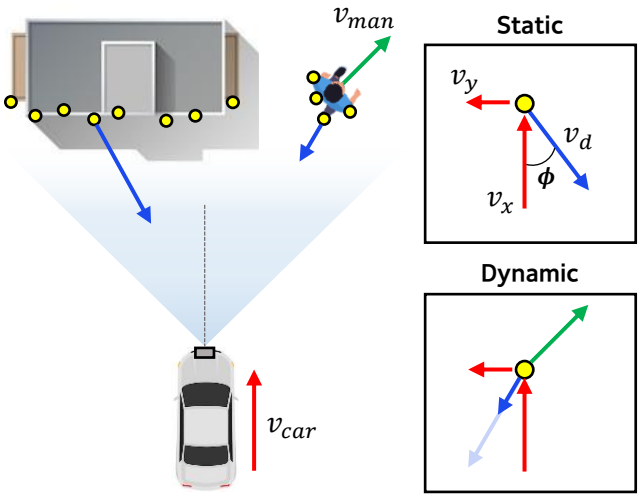


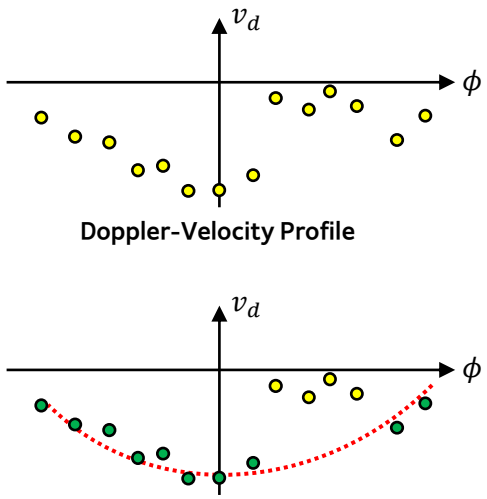
Fig 10. CFAR result using a CA threshold (α = 5, guard = 1, window = 10)

- In a noise environment, very weak echo signals may be lost rather than in the case of a fixed threshold.
- CA-CFAR also has limitations in that performance is guaranteed only in uniformed noise environments.
  - GO(Greatest-of) CFAR, LO(Least-Of) and others also adaptive method. It depends on what targets want to detect

RADAR FILTERING METHOD – DYNAMIC OBJECT

- Dynamic object filtering helps radar systems distinguish between actual targets and non-target objects.
  - Moving objects such as aircraft in flight, and vehicles are dynamic objects that reflect radar signals
- The Doppler-Velocity profile is a commonly used and Intuitive way to differentiate static/dynamic objects
  - Doppler-Velocity value depends on not only the direction between radar and objects but also object speed.
  - If the azimuth between the target and radar is bigger, the doppler-velocity value is smaller. But dynamic objects deviate from this rule.
- Apply Curve-Fitting to profile, such as the RANSAC algorithm, which makes It easier to distinguish between dynamic and static objects from the returned targets.
  - It is only applicable in environments mostly composed of static objects.





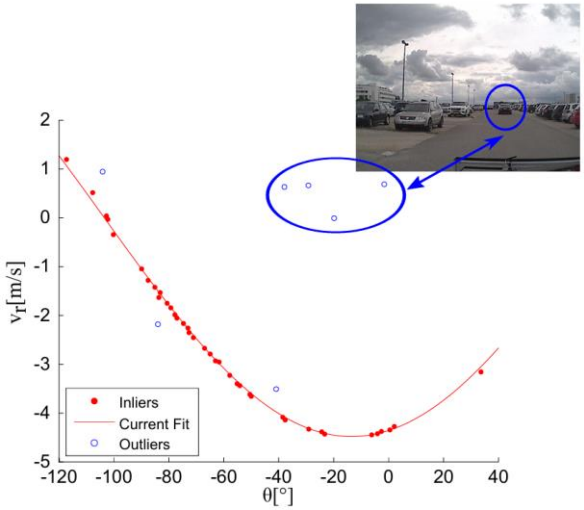
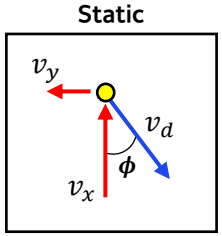


Fig 11. Dynamic object filtering result

RADAR FILTERING METHOD – VELOCITY

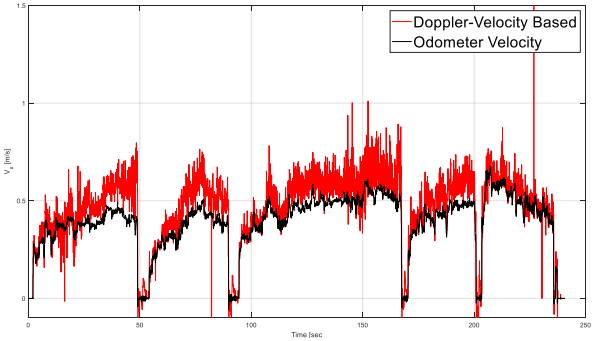
- The Doppler velocity obtained from the radar, quite literally, the Doppler velocity.
  - If radar is mounted to the vehicle, doppler velocity is not the actual vehicle velocity.
  - In the field of navigation, what is needed when using radar is the speed of the vehicle.
- From the velocity profile, one can derive the velocity equation of the vehicle.



$$-v_d = v_x \cos\phi + v_y \sin\phi$$

- Even though dynamic objects are distinguished, there still exists Doppler-velocity for many static objects.
  - This doppler-velocity measurement may involve noise and interference.
- The most commonly used estimation algorithm 'Least-Squares' method to estimate the vehicle velocity from Doppler velocity.
  - The limitation of this method is that its accuracy depends on the performance of the radar.

$$-\boldsymbol{v}_d = \begin{bmatrix} \cos\phi_1 & \sin\phi_1 \\ \cos\phi_2 & \sin\phi_2 \\ \vdots & \vdots \\ \cos\phi_n & \sin\phi_n \end{bmatrix} \begin{bmatrix} v_x \\ v_y \end{bmatrix}$$



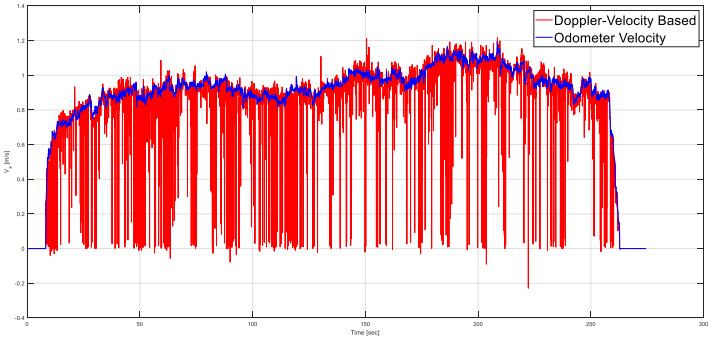
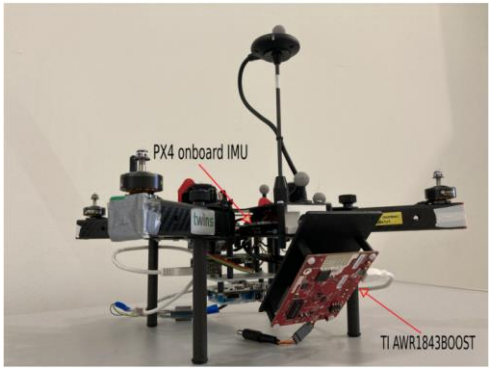


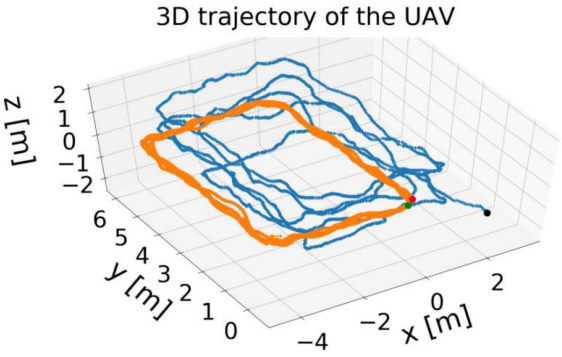
Fig 12. Velocity Estimation result (UMRR 11-type 132 and TI IWR6843ISK)

mmWave Radar for Navigation - Localization

- Tightly-coupled EKF-Based Radar-Inertial Odometry [2]
  - IMU is a core sensor used for the system state propagation, Updates are performed with radar measurement.
  - Using stochastic cloning of past robot poses for formulating an update equation.
  - Matching 3D points across sparse, noisy radar scans in full 6DoF motion allowing correspondence generation for point-distance-based updates with a single past radar pose.





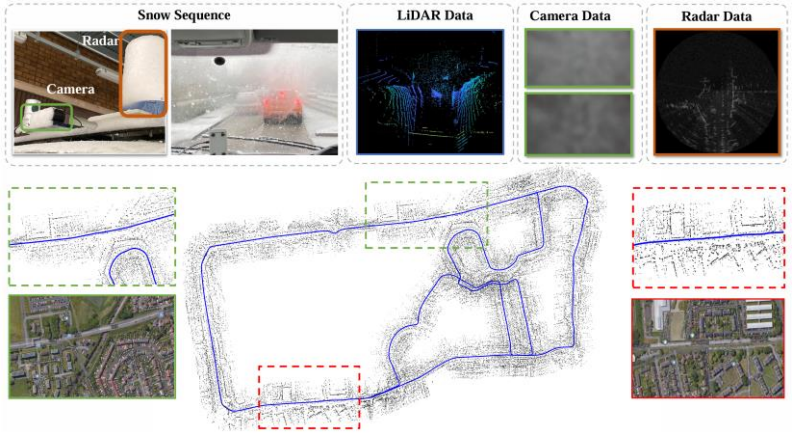


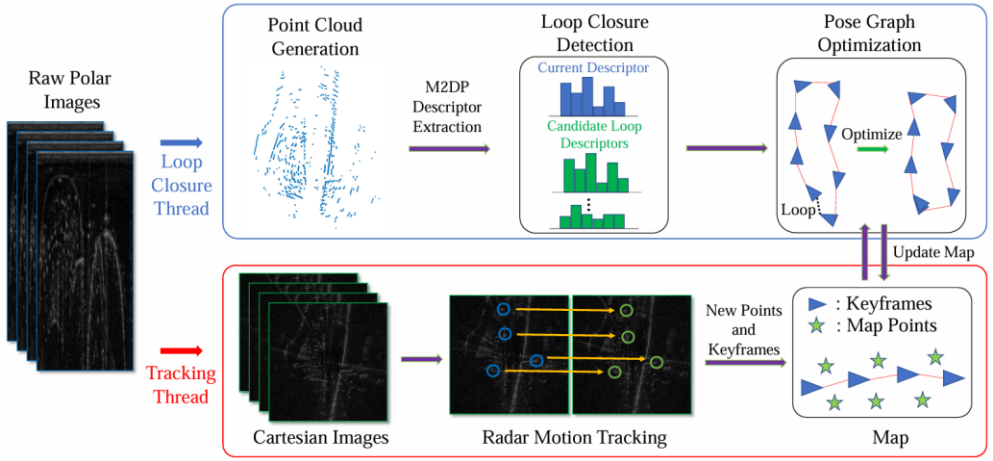
- Using both distance from **radar** scans and velocity, Accomplish accurate 6D pose and 3D velocity estimation of the mobile platform.

[2] Michalczyk, Jan, Roland Jung, and Stephan Weiss. "Tightly-coupled ekf-based radar-inertial odometry." *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2022.

mmWave Radar for Navigation - Mapping

- Radar SLAM : A Robust SLAM System for All Weather Conditions [3]
  - Extensive experiments on extreme weather conditions (Snow, Fog/Rain, Night).
  - Robust data association and outlier rejection mechanism for radar-based feature tracking by leveraging radar geometry.
  - Fomulating motion compensation model to reduce motion distortion induced by a low scanning rate.





- Without additional sensors, the scanning radar alone can be used to perform localization and mapping
- Radar has Superiority and promising potential and It can be the primary sensor for long-term localization and navigation tasks.

[3] Hong, Ziyang, et al. "Radar SLAM: A robust SLAM system for all weather conditions." *arXiv preprint arXiv:2104.05347* (2021).

SUMMARY

- To properly utilize mmWave radar systems, I surveyed the basic principles of radar and filtering methods to how they are applied in navigation systems.
- mmWave radar has different characteristics compared to conventional pulse radar, and its use of short wavelengths allows it to obtain precise measurements from various objects.
- Radar has become a core robotics sensor in applications for operating in visually degraded environments. As the reliability and accuracy of radar sensors increase and their prices decrease, the prevalence of these sensors will continue to grow.

Q&A

